

|                  |                          |                |                           |
|------------------|--------------------------|----------------|---------------------------|
| INCIDENT ID      | INC-84920                | INCIDENT LEAD  | Alex Mercer (DevOps Lead) |
| SERVICE IMPACTED | Web Portal & API Gateway | START TIME     | July 23, 2026 – 14:12 UTC |
| CURRENT STATUS   | RESOLVED                 | RESOLVED TIME  | July 23, 2026 – 14:54 UTC |
| IMPACTED USERS   | ~38% Active Sessions     | TOTAL DURATION | 42 Minutes                |

## EXECUTIVE SUMMARY

### High-Level Overview

At 14:12 UTC, the primary customer portal experienced a steep spike in HTTP 504 Gateway Timeouts, causing active users to be logged out and preventing checkout completions. The incident was traced to an unindexed database query introduced in deployment release v2.14.0, which exhausted the PostgreSQL database connection pool under peak traffic. Full service functionality was restored after initiating a build rollback and executing an emergency database indexing patch.

## INCIDENT TIMELINE

| TIME (UTC) | EVENT / STAGE          | DESCRIPTION                                                                                                                            |
|------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 14:00 UTC  | Deployment Released    | Release v2.14.0 (User Analytics Module) deployed to Production cluster via automated CI/CD pipeline.                                   |
| 14:14 UTC  | Alert Triggered        | Datadog latency monitor triggered PagerDuty alert: 95th percentile latency exceeding 4,500ms and HTTP 500/504 error rate spikes > 15%. |
| 14:18 UTC  | Incident Bridge Opened | On-call Incident Commander opened war room bridge. Infrastructure, DB, and Backend leads joined triage.                                |
| 14:32 UTC  | Root Cause Identified  | Database Administrator identified query lockup on users_login_activity table due to missing index on last_login_timestamp.             |
| 14:40 UTC  | Rollback Initiated     | Initiated automated rollback from v2.14.0 back to stable build v2.13.9. Blocked DB connections cleared manually.                       |
| 14:54 UTC  | Full Service Recovery  | Traffic metrics returned to baseline. CPU utilization dropped below 25%. Synthetic testing verified checkout and authentication flows. |

## ROOT CAUSE ANALYSIS (RCA)

**Primary Cause:** Release v2.14.0 included an active user tracking feature that introduced a database migration lacking an index on the last\_login\_timestamp column. **Contributing Factor:** In the pre-production staging environment, low record volume kept query times under 15ms. Under production load (10,000+ requests/sec), query execution degraded to ~3,800ms, rapidly saturating the database connection pool. **Detection & Prevention Gap:** Synthetic load testing scripts did not accurately simulate full production scale database sizes or concurrent connection spikes.

## CORRECTIVE & PREVENTATIVE ACTIONS

| ACTION ITEM                                                                            | OWNER           | TARGET DATE  | STATUS      |
|----------------------------------------------------------------------------------------|-----------------|--------------|-------------|
| Apply missing index patch to users_login_activity table                                | DBA Team        | Immediate    | COMPLETED   |
| Update staging database scale to replicate production data volumes                     | QA / DevOps     | Aug 02, 2026 | IN PROGRESS |
| Implement CI/CD linter rule to flag missing foreign key & lookup indices in migrations | DevOps / SecOps | Aug 10, 2026 | PLANNED     |
| Tune auto-scaling and connection pooling limits for API Gateway                        | Backend Team    | Aug 15, 2026 | PLANNED     |